

TrES-2b

R-2406

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_120611 · created 2026-08-21T00:54:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456089.77507 +/- 0.00016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.019
Transit depth (Rp/Rs)^2	1.28 +/- 0.42 [%]
Orbital Inclination (inc)	86.34 +/- 1.65
Airmass coefficient 1 (a1)	0.989 +/- 0.011
Airmass coefficient 2 (a2)	0.008 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#2 - [28, 118]
Optimal Aperture	4.08
Optimal Annulus	12.7
Transit Duration (day)	0.095 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.113 ± 0.019 vs 0.1254 (deviation 0.65σ)
Duration fit vs published	0.095 d vs 0.0746 d (deviation 1.28σ)
Mid-time O–C	0.3 min
Depth SNR	5.9
Residual scatter	1.11 %

Light curve

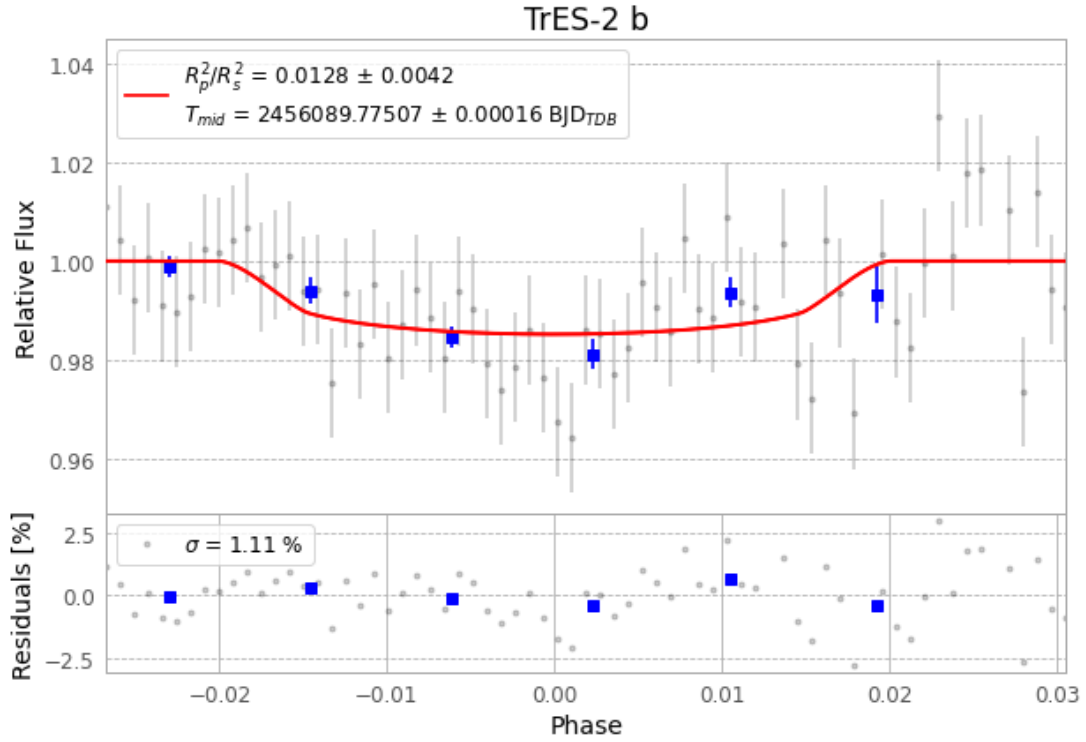


Figure 1 — FinalLightCurve_TrES-2 b_2012-06-10.png (SHA-256 da48d57e7851ae13...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 198], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d743ed9c8fd4976e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), TRES-2120611045712.FITS → TRES-2120611082114.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-120611.log (2b8e762fc0ac...), FinalLightCurve_TrES-2 b_2012-06-10.png (da48d57e7851...), FinalLightCurve_TrES-2 b_2012-06-10.csv (f2cf403f72f5...)

Compute resources

Wall time 102.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:54Z.